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Biochar for durable carbon removal: soil erosion reduction as a key mechanism

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ABSTRACT

Soil erosion is a major pathway of physical soil organic carbon (SOC) loss and a critical threat to the permanence of land-based carbon dioxide removal (CDR). Biochar is widely recognised as a durable carbon sink, yet its hydrological effects and erosion-mitigation potential remain undervalued in CDR frameworks. Here we synthesise global evidence and long-term Mediterranean experiments to show that hydrological improvements—an enhanced soil sponge function—are not ancillary co-benefits but a core mechanism supporting carbon durability. A global meta-analysis indicates that biochar reduces runoff by 25 % and soil erosion by 16 % on average. In Mediterranean vineyards, we observed up to 65 % soil erosion rate reduction (fine-earth and coarse fragment (gravel) erosion), up to 73 % increases in stored water (up to 300 % in dry conditions), and 85 % increases in SOC with 4 % (w/w) biochar amendments. These findings imply that soil erosion reduction should be formally integrated into Monitoring, Reporting, and Verification (MRV) methodologies and/or considered in forthcoming IPCC methodology reports on CDR, carbon capture and storage (CCS), and carbon capture and utilisation (CCU) for biochar-based CDR. Explicitly accounting for erosion-mediated carbon retention would reduce permanence uncertainty and strengthen the credibility of soil-based CDR credits in emerging compliance and voluntary carbon markets. By protecting carbon stocks in topsoils and enhancing crop and vegetation productivity, biochar provides a scalable strategy for climate-resilient land management and durable carbon removal with the added benefit of preserving existing soil C stocks.

1. Introduction

Soil erosion is one of the largest unaccounted pathways for the loss of soil organic carbon (SOC) [1], undermining global climate change mitigation and land restoration efforts [2]. The IPCC [3] highlights that erosion-driven SOC export directly threatens the permanence of land-based carbon dioxide removal (CDR) strategies (Fig. 1 provides an overview of CDR methods). Across Europe, erosion hotspots—particularly Mediterranean landscapes—lose SOC at rates far exceeding natural replenishment [4]. LUCAS Soil Survey data indicate that many Mediterranean soils contain critically low SOC stocks, often below 1 % [5], making them extremely vulnerable to hydrological extremes and climate-driven degradation.

Biochar application is widely recognised for its potential to increase SOC stocks [7], improve soil structure, and enhance long-term carbon storage [8]. An underexplored dimension is how biochar's hydrological effects—such as increasing infiltration and reducing runoff—affect erosion-driven losses of SOC. These hydrological effects do not directly create carbon removal, but they help preserve newly accumulated SOC derived from increased plant productivity and other regenerative practices. By reducing the physical export of this additional CDR-relevant SOC, improved hydrological function strengthens the permanence of land-based carbon sequestration following biochar application.

Recent policy developments, including the EU Carbon Removal Certification Framework (CRCF) [9], emphasise the need for verifiable permanence metrics and robust risk mitigation for soil-based removals. Yet present methodologies do not explicitly account for erosion reduction as a measurable carbon-retention mechanism. Emerging policy analysis further stresses that the climate relevance of biochar depends on century-scale persistence supported by registered material properties, conservative modelling, and long-term field validation, rather than assumptions of near-complete inertness over millennial timescales [10].

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Therefore, we aim to evaluate whether biochar-induced improvements in soil hydrology—specifically increased infiltration, reduced runoff, and lower erosion—can be quantified as mechanisms that enhance the retention of newly formed SOC and thereby strengthen the permanence of land-based CDR.

Recent modelling work on Enhanced Rock Weathering (ERW) highlights its potential as a complementary land-based CDR pathway, but assessments have focused almost exclusively on geochemical carbon fluxes [11]. To date, there are no field studies that directly quantify soil erosion and runoff responses to ERW applications, leaving a critical gap in understanding how physical soil processes may affect carbon permanence. Preliminary evidence indicates that crushed rock amendments can improve aggregate stability and hydraulic conductivity [12], suggesting that ERW may influence erosion susceptibility and SOC retention through physical pathways. Although ERW is not directly comparable to biochar-based CDR, this gap underscores the broader relevance of erosion and hydrological controls for evaluating permanence across land-based (and geochemical) CDR strategies. This parallel underscores our central point: hydrological controls condition the real-world permanence of land-based CDR.

2. Global evidence: hydrology as a driver of carbon durability

A global meta-analysis covering 184 observations from 30 studies (Fig. 2) shows that biochar reduces runoff by 25 % and soil erosion by 16 % (global mean). The variation across Temperate, Subtropics, and Tropics climatic zones is shown in Fig. 3 [13]. At the experimental scale, 47 % of the tests were conducted in the field, while 53 % were carried out in the lab or greenhouse. Erosion mitigation was most effective at topsoil concentrations ranging from 0.6 % to 2.5 % by weight (w/w). The effectiveness doubled when biochar was pyrolysed at temperatures above 500 °C, which is consistent with a higher carbon content, a larger internal surface area and porosity and greater physical stability of the material. About 31 % of biochars were wood-derived.

These findings indicate that the hydrological effects of biochar—enhanced infiltration and water storage, reduced bulk density, and improved aggregate stability—are central to its CDR potential. The interaction with vegetation is particularly relevant. Studies with plant cover show more than twice the erosion reduction as compared to bare soil experiments, suggesting a strong synergy between biochar's soil physical changes and enhanced above-ground biomass production.

3. European validation: long-term evidence

Mediterranean region, particularly vineyard soils, which are mainly medium-coarse in texture, often unirrigated and located on slopes, represent some of the most erosion-prone agricultural landscapes in Europe [14]. Wu et al. [15] have shown that medium-coarse-textured soils have a greater potential to reduce runoff and erosion compared to fine-textured soils. To draw robust conclusions about the effects of biochar on soil sponge function and soil erosion, Gholamahmadi et al. [13] recommend conducting long-term experiments on these soil textures. Soil sponge function refers to the capacity of soil to absorb, retain, and release water and nutrients, while simultaneously providing structural support to resist erosion and maintain SOC stocks.

The experiments were conducted over a period of 30 months (18

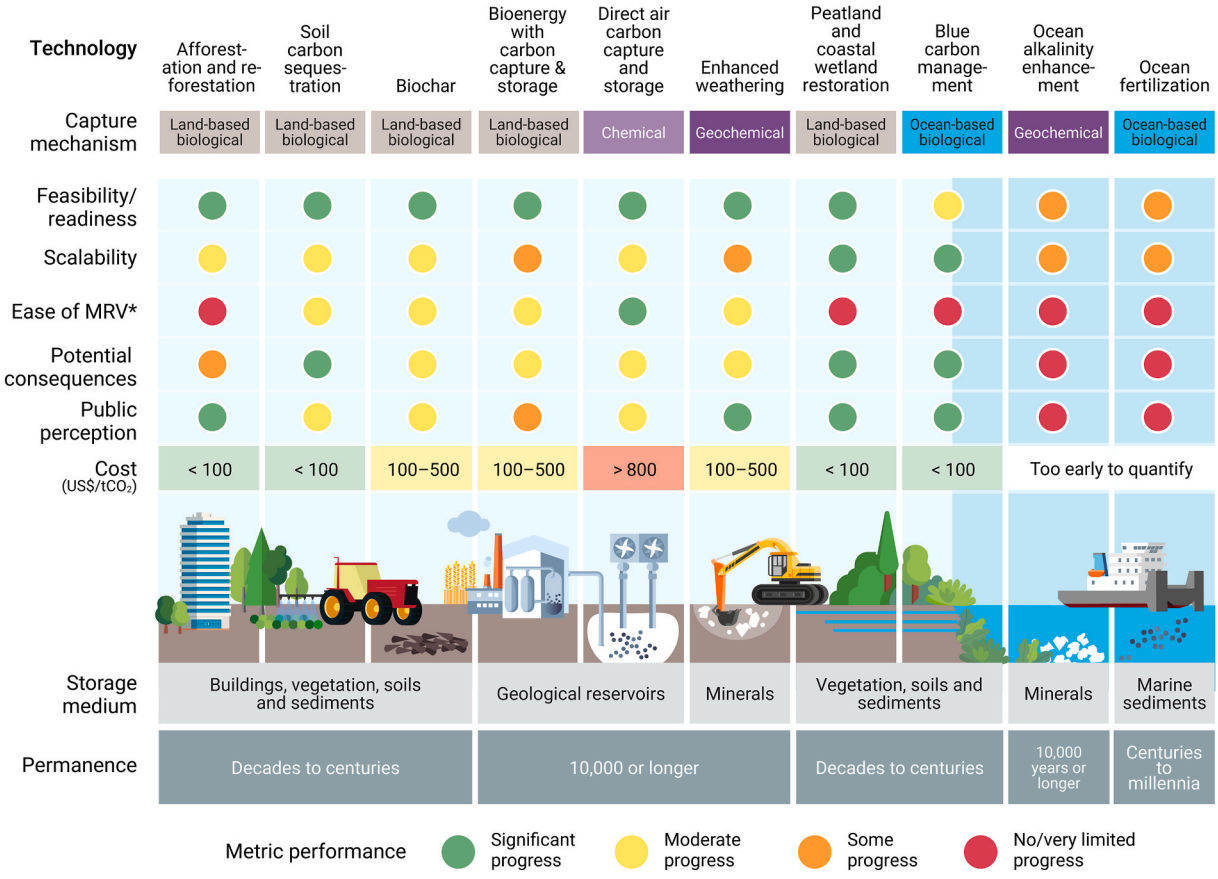


Fig. 1. Illustrative overview of CDR methods and their main characteristics [6]. * Monitoring, Reporting and Verification. Ease of MRV indicates the level of development in practices for CO_2 monitoring and accounting for a given CDR option. The coloured circles indicate the level of progress for different metrics based on current technology development: green means progress is close to target levels required for wider adoption, whereas red indicates no progress or only limited progress toward the target. For example, green indicates that MRV is well-developed, with standardised methodologies and techniques supported by comprehensive demonstration trials. At the same time, red corresponds to either a lack of practical experience or technical difficulties in MRV development.

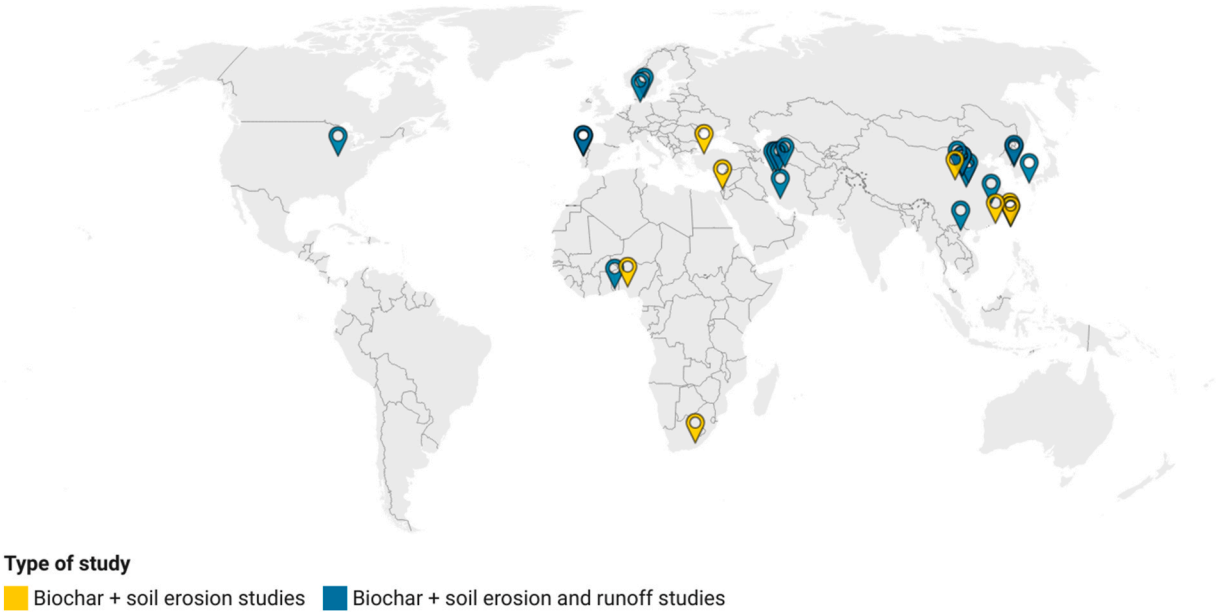


Fig. 2. Illustrative global distribution of study sites from the published meta-analysis [13]. Visualisation URL: https://datawrapper.dwcdn.net/gGvaW/1/.

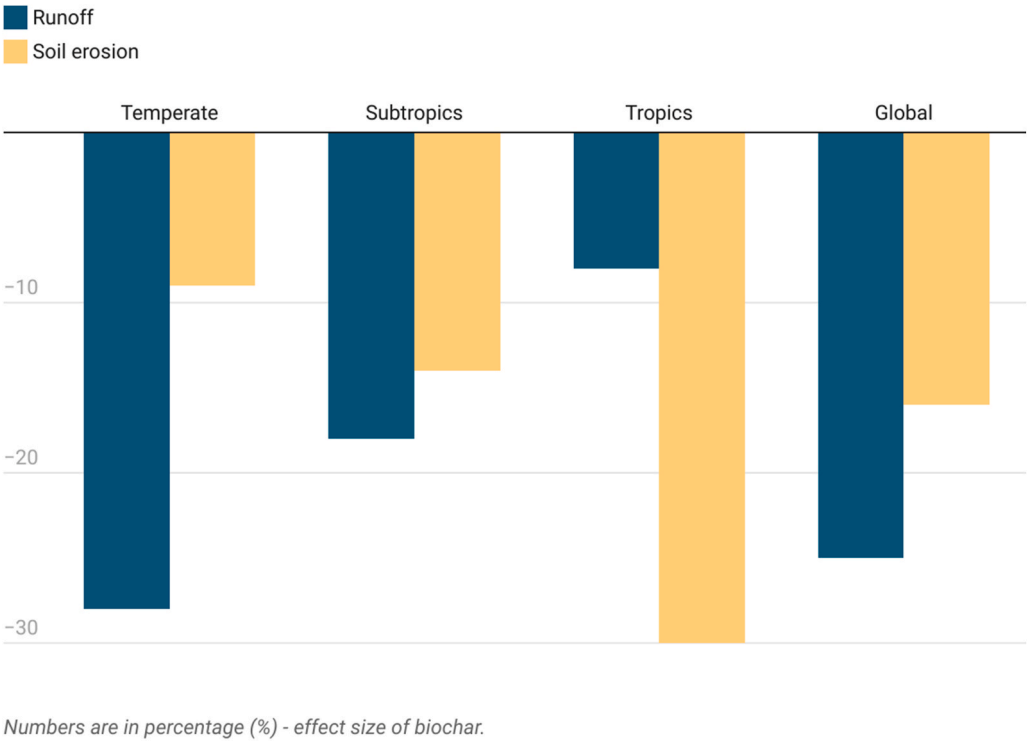


Fig. 3. Illustrative effect size of biochar on runoff and soil erosion by water across three climatic zones—Temperate, Subtropics and Tropics—compared to the global mean [13]. Visualisation URL: https://datawrapper.dwcdn.net/AtL3K/1/.

months and 12 months), using an outdoor lysimeter field in a random block design ($n = 3$) under natural rainfall conditions. The main focus was to evaluate the soil sponge function and soil erosion. First experiment (2016–2018) showed 4 % (w/w) biochar addition to sandy loam soil reduced runoff by 45 % and decreased erosion rate by 65 %, and by 34–67 % depending on fraction size (splash erosion < fine-earth erosion < coarse fragment (gravel) erosion). Stored water increased by 73 % on average, causing up to 300 % more retained water during dry periods [16].

In another experiment from 2019 to 2020, biochar reduced erosion by 58 % (fine-earth fragments), increased infiltration by 50 %, and enhanced water retention by 59 %. Vegetation cover increased by 263 %, driven by improved soil pH, NPK availability, and water retention [17]. Biochar application, in combination with vegetation cover, reduced the annual erosion rate (fine earth fragments) from 2.05 t ha^{-1} y^{-1} to 0.76 t ha^{-1} yr^{-1}. This was substantially lower than the tolerable soil erosion rate (i.e. 1–2 ha^{-1} yr^{-1}) reported in European agricultural soils [18].

The enhancement of hydrological properties in the medium-textured soils confirmed the findings of the meta-analysis conducted by Wei et al.

[19]. These changes were accompanied by substantial SOC increases (up to 85 %) relative to the control. Because SOC was measured as total organic carbon, the rise reflects both added biochar-C and any change in native SOC (i.e., this is not a pyrogenic vs non-pyrogenic partitioning). Under plausible assumptions, the measured SOC increase (11–33 t C ha^{-1}) remains below the amount of biochar-C applied (≈36 t C ha^{-1}), indicating that the observed SOC gains are consistent with the added biochar being the main driver, with only a minor contribution from native or newly formed SOC.

The cited studies are representative reference points: the global meta-analysis provides the first quantitative synthesis of biochar effects on runoff and soil erosion, while the accompanying long-term field studies serve as independent validation under European agroecosystem conditions. The full spatial distribution of studies included in the analysis is accessible via the interactive map linked in Fig. 2. Also, detailed descriptions of experimental design and control of confounding factors in Mediterranean studies are provided in the original publications.

4. Mechanistic perspective: why can hydrological functions enter CDR MRV?

Current biochar CDR methodologies primarily quantify carbon storage based on the stability of the pyrolysed carbon fraction and modelled mineralisation rates. However, topsoil loss through erosion represents an immediate reversal threat, exporting both biochar carbon and newly accumulated SOC from the system, thereby reducing or even offsetting other (non-biochar-related) soil C sink activities and measures as described by the 4per1000 initiative [20].

Hydrological function is therefore a central mechanism for i) Carbon retention: Reduced runoff prevents the lateral export of SOC, ii) Carbon accrual: Higher water availability increases primary productivity and organic matter inputs, iii) Soil stability: Improved aggregation reduces susceptibility to raindrop impact and surface crusting.

The mechanisms outlined above are supported by a global meta-analysis of biochar impacts on runoff and soil erosion [13], as well as long-term European field studies [16,17]. Together, these studies provide empirical evidence that enhanced hydrological function is not merely an ancillary benefit but a central pathway for carbon retention, accrual, and soil stability. These mechanisms should be further investigated and, where well understood, integrated into future MRV protocols, particularly under the CRCF, to quantify additional C-sinks and permanence associated with hydrological improvements.

Hydrological improvements interact with other biochar stabilisation mechanisms. By reducing runoff and enhancing soil water retention, they protect both native and added SOC from physical loss, while creating favourable conditions for mineral association and the long-term persistence of chemically recalcitrant carbon [21,22]. These synergistic effects underscore the uniqueness of the hydrological mechanism as a complementary pathway for durable carbon sequestration. Potential trade-offs, such as particle loss or differences in soil texture, may influence the relative contributions of each mechanism and should be considered in future research and MRV protocols [23]. While some near-surface particle redistribution can occur, net soil erosion reductions dominate at the field scale.

5. An open research question: hydrological and structural pathways for SOC persistence

Published literature indicates that, under certain settings, biochar can increase non-pyrogenic SOC in systems managed for SOC accrual (e.g., no-till, grasslands), but findings are mixed. Some studies report additive gains linked to negative priming, increased microbial carbon use efficiency, and stabilisation of microbial necromass in biochar pores—implying reduced mineralisation of native SOC under biochar amendments [24]. Conversely, observations in subtropical soils show occasional positive priming effects that can decrease native SOC, highlighting that additive gains are not universal and may depend on soil texture, rainfall regime, and biochar characteristics [25]. Hydrological mechanisms remain a leading hypothesis: meta-analyses show that biochar often enhances soil water retention, aggregate stability, and reduces runoff and erosion, potentially protecting native and newly formed SOC from physical export [13,26,27]. These effects are strongly influenced by soil texture, biochar porosity, and application rate, with coarser textured soils often exhibiting pronounced improvements in water-holding capacity and infiltration [17,19,28]. Structural mechanisms include enhanced aggregate stability and increased formation of mineral-associated carbon fractions, which promote SOC protection, with effects varying by pyrolysis conditions and soil chemistry [29]. Management interactions also matter; for example, biochar applications under conservation tillage and in perennial grassland systems have been associated with increases in native SOC and a greater recalcitrant fraction, suggesting that practice-specific synergies can favour non-pyrogenic SOC accrual [17,30]. Finally, analytical and policy frameworks caution that routine SOC measurements [31] cannot reliably distinguish pyrogenic from non-pyrogenic carbon, reinforcing the need for multi-site, long-term experiments to determine: (a) whether hydrological effects contribute to additive SOC increases, and (b) under which soil and ecosystem conditions these effects occur and can be credibly incorporated into MRV protocols.

6. Implications for biochar-based deployment

Six implications arise for climate policy and land management.

i. Although erosion reduction is typically classified as an emissions-reduction strategy rather than a permanence mechanism per se, its hydrological effects directly support carbon retention and should therefore be incorporated into carbon-removal accounting frameworks.
ii. Mediterranean and other low-SOC, erosion-prone regions offer high additionality. The relative improvements—up to 65 % erosion reduction and greater than 70 % increase in water storage—far exceed global averages.
iii. Quantifying erosion-related carbon retention can be approached using SOC content of the topsoil combined with measured or modelled erosion rates, providing a first-order estimate of C retention per unit soil loss. Empirical evidence from Mediterranean field (outdoor lysimeter) trials and global meta-analysis indicates that biochar amendments can reduce erosion substantially, which can be translated into an erosion-modified SOC retention factor for use in MRV models. While site-specific calibration is required, this provides a feasible pathway for incorporating hydrological improvements into carbon accounting frameworks. This supports an ‘erosion-modified SOC retention factor’ for MRV, with site-specific calibration.
iv. Biochar–vegetation synergies enhance long-term ecosystem resilience. Carbon removal is strengthened by cascading co-benefits: higher pH, improved nutrient cycling, increased biomass, and soil regeneration.
v. Large agricultural systems such as those in China, where soil erosion and SOC stabilisation are longstanding priorities, may particularly benefit from integrating erosion-related permanence considerations into biochar deployment and certification frameworks.
vi. Biochar application in tropical and subtropical systems, particularly across the Global South, may also offer benefits, but site-specific constraints (e.g., high rainfall intensity, rapid SOC turnover, diverse soil type, and feedstock availability) must be considered, and additional field studies are needed to support MRV frameworks in these regions.

7. Limitation and applicability

The cost and therefore the market price of sustainably produced biochar vary widely depending on feedstock availability, production technology, scale, and system integration [32]. While economics are critical for scale, a comprehensive cost assessment is outside our scope; we focus on biophysical mechanisms of SOC permanence. Importantly, the hydrological benefits documented here may improve cost-effectiveness indirectly by reducing SOC losses, enhancing productivity, and increasing the durability of carbon removal outcomes.

Field and meta-analysis evidence indicate that, despite some physical redistribution, net soil erosion is reduced, thereby protecting both native SOC and applied biochar. Biochar-C is highly recalcitrant (decades to centuries), and hydrological improvements further enhance its durability.

8. Conclusion

Biochar's ability to enhance infiltration, increase soil water retention, and reduce erosion, thereby limiting physical SOC export, directly supports the permanence of existing land-based carbon stocks while adding carbon dioxide removal via biochar. Global and Mediterranean evidence shows that hydrological improvements are a key mechanism underpinning long-term carbon storage and ecosystem resilience. Integrating erosion-mitigation metrics into CDR MRV frameworks or international emission and CDR inventories following the forthcoming IPCC methodologies will improve the robustness, credibility, and scalability of biochar as a durable climate solution. Recognising erosion mitigation within MRV can accelerate credible deployment of biochar as a durable CDR pathway in erosion-prone agroecosystems.

CRediT authorship contribution statement

Behrouz Gholamahmadi: Writing – original draft, Resources, Methodology, Investigation, Conceptualization. **Claudia Kammann:** Writing – review & editing, Validation.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used AI-assisted tools, including Microsoft Copilot, ChatGPT (OpenAI), and Grammarly, to support language editing, grammar checking, readability improvement, and formatting of references (e.g., APA style). These tools were not used to generate scientific content, data, analyses, or interpretations. After using these tools, the authors critically reviewed and edited the manuscript and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No new data were generated or analysed in this study. All data discussed in this Perspective are derived from previously published studies and publicly available sources cited in the reference list.

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